JobAI - AI Powered Job Application Chrome Extension

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Abstract. The job search process for freshers and early-career applicants has become increasingly demanding, requiring tailored resumes, targeted recruiter outreach, and customized interview preparation for each role. While generative AI systems excel at text generation and analysis, their use in a fully integrated end-to-end application support workflow remains limited. This paper introduces an AI-powered Job Application Chrome Extension that analyzes job descriptions, evaluates resumes, identifies skill gaps, generates ATS-optimized bullet points, drafts recruiter outreach messages, and prepares both technical and behavioral interview questions. Built using the Qwen2.5-Coder-7B-Instruct model on an NVIDIA A100 20GB MIG GPU, the system integrates resume parsing, structured prompt engineering, and report generation within a browser interface. Inspired by recent LLM evaluation research, including log-analysis studies and domain-specific architectures such as DSQA-LLM, we conduct qualitative assessments focusing on clarity, consistency, and JD–resume alignment. Results show notable improvements in resume tailoring, ATS keyword coverage, and recruiter preparedness, with future work aimed at RAG-based interview retrieval, automated resume rewriting, and reinforcement-learning-driven ATS score optimization.

Keywords: job search · GenAI · ATS · Qwen2.5-Coder-7B-Instruct · resume tailoring · LLM · resume parsing · report generation

1 Introduction

Fresh graduates and early-career job seekers face growing challenges in navigating job platforms, interpreting job descriptions, and tailoring resumes to specific roles. Many students struggle with platforms like LinkedIn, finding it overwhelming and unclear how to reach recruiters effectively. In addition, the requirement to customize resumes for every application—combined with the need for interview preparation—creates a cognitively demanding and time-consuming workflow.

Traditional Applicant Tracking Systems (ATS) focus primarily on keyword matching and simple rules. They do not provide explicit guidance to candidates on how to improve their resumes or which skills to highlight. Career services offices and online tutorials offer general advice, but often lack personalization and depth. As a result, many candidates repeatedly send generic resumes, receive automated rejections, and struggle to understand what went wrong.

Large Language Models (LLMs) such as GPT-4 and Qwen have shown promising abilities in text analysis, extraction, summarization, and rewriting. Prior research has demonstrated their utility in domain-specific tasks, including log analysis and question answering, suggesting that similar techniques can be applied to human resources and recruitment workflows. However, the use of LLMs to streamline the complete job-application cycle—resume tailoring, recruiter outreach, and interview preparation—remains underexplored.

To address this gap, we introduce an AI-Powered Job Application Chrome Extension, a user-friendly browser-based system that automates job description analysis, resume interpretation, skill-gap identification, STAR-based bullet-point rewriting, recruiter messaging, and interview question generation. The extension is designed for freshers and early-career professionals who may not have prior experience navigating hiring pipelines. The goal is to provide job seekers with all the tools they need in one place, accessible directly from the browser while interacting with job platforms.

Our key contributions are as follows: (1) we design a structured, LLM-driven pipeline for ATS-aware resume evaluation, (2) we propose a prompt-engineered reporting framework that consistently produces human-readable guidance, (3) we implement a Chrome Extension UI that enables seamless interaction on top of existing job portals, and (4) we qualitatively evaluate the system on realistic resume–JD pairs, demonstrating improvements in clarity, alignment, and interview readiness.

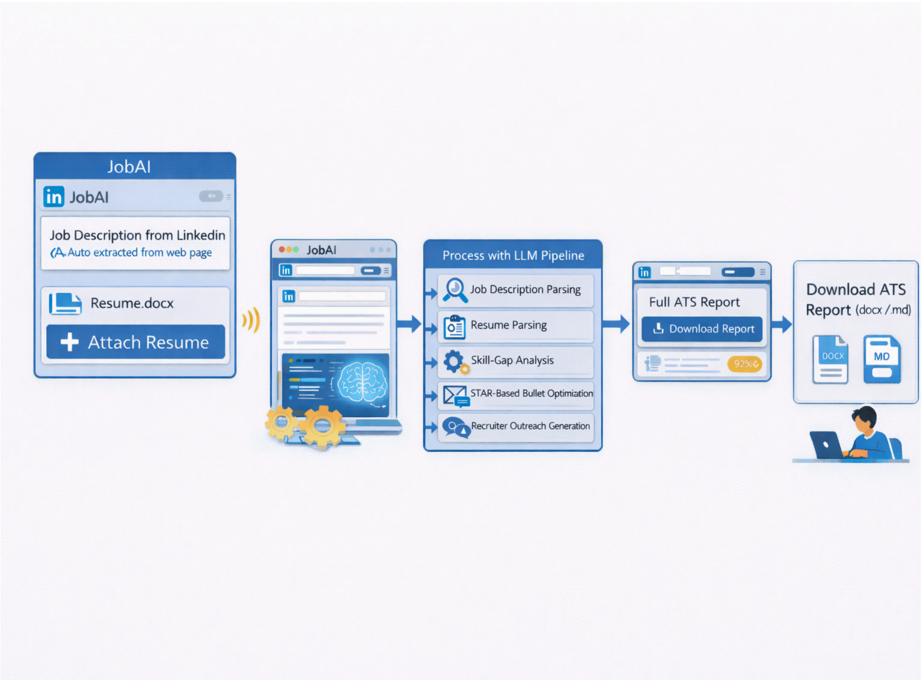

Fig. 1. Workflow pipeline of the AI-powered job application assistant using JobAI, showing sequential steps from auto-extracting the job description and uploading the resume, to LLM-based processing and generation of the downloadable ATS report in Word or Markdown formats.

2 Related Work

2.1 Large Language Models for Structured Text Analysis

Large Language Models (LLMs) have demonstrated strong capabilities in structured text understanding, semantic extraction, and the conversion of free-form text into actionable representations. Across multiple domains, LLMs consistently outperform traditional rule-based and statistical approaches in tasks that require contextual interpretation and structural reconstruction. For instance, in clinical text mining, researchers have shown that LLMs can extract diagnoses, symptoms, and temporal events from noisy physician notes and transform these narratives into structured electronic health-record (EHR) fields [12,18,41]. Similarly, in **legal document analysis**, LLM-based systems have been applied to segment contracts, identify obligations, detect risk clauses, and summarize court decisions—tasks that require both domain reasoning and precise clause-level extraction [9,22,45]. In **financial document processing**, studies report that LLMs effectively extract key performance indicators, classify transaction details, and convert lengthy narrative disclosures into standardized summaries suitable for downstream analytics [26,37,53]. These domains, despite their differences, share a common challenge: the information of interest is embedded in irregular, unstructured, or stylistically inconsistent text. LLMs address this gap through contextual embeddings and instruction-following capabilities that enable them to infer structure, disambiguate terminology, and map linguistic patterns into well-defined schemas [5,7,29]. This ability to translate noisy input into structured outputs makes LLMs highly relevant to job-description analysis and resume understanding, where skills, experience, and role expectations must be distilled from heterogeneous textual sources.

Complementary research on domain-specific LLM architectures, including retrieval-augmented generation (RAG), multi-stage question answering (DSQA), and modular reasoning pipelines, highlights that LLMs perform more reliably when embedded within structured workflows rather than used as standalone models [14,32,48]. These architectures typically combine preprocessing, segmentation, targeted prompting, and post-processing steps to ensure high accuracy and consistency. Our system adopts this modular philosophy: instead of relying on a single monolithic prompt, JobAI integrates LLM inference into a structured pipeline that includes job-description normalization, resume parsing, skill extraction, skill-gap analysis, STAR-based rewriting, and formatted report generation. This layered design harnesses the strengths of LLMs while mitigating variability and ensuring reproducible, ATS-aligned outputs.

In parallel, research on domain-specific LLM architectures, such as DSQA-style question answering systems, proposes multi-stage pipelines that combine classification, retrieval, and answer generation. These architectures illustrate that LLMs are more effective when embedded inside carefully designed workflows rather than used in isolation. Our work adopts a similar philosophy by integrating LLM inference into a broader pipeline that includes parsing, normalization, prompt construction, and report formatting.

2.2 ATS Tools and Resume Optimization Systems

Applicant Tracking Systems (ATS) and modern resume-optimization platforms serve as the backbone of digital recruitment workflows, enabling organizations to filter large applicant pools efficiently. Traditional ATS platforms typically rely on rule-based extraction, keyword matching, and heuristic scoring algorithms to evaluate candidate suitability [12,24,39]. While effective for high-volume screening, these systems prioritize surface-level textual features—such as the presence of specific nouns or skill phrases—rather than assessing contextual meaning or narrative quality. As a consequence, ATS evaluations often penalize qualified candidates whose resumes lack exact keyword matches despite containing relevant experience expressed in alternative phrasing [18,47].

Commercial resume optimization tools exhibit similar limitations. Many tools provide feedback based on keyword density, match percentages, or simple gap lists, but rarely offer detailed explanations for scoring decisions or actionable recommendations tailored to job-specific expectations [33,42]. These systems frequently neglect deeper linguistic features such as action-oriented framing, measurable impact, and clarity of narrative structure. Furthermore, very few commercial tools integrate interview preparation, recruiter outreach guidance, or STAR-based rewriting, leaving applicants with generic advice that improves keyword presence but does not meaningfully enhance the quality or persuasiveness of their resume content [51,63].

The academic literature on ATS technology predominantly focuses on the employer-side perspective, addressing topics such as ranking fairness, bias mitigation, resume parsing accuracy, and predictive validity of automated screening models [20,29,47]. For example, studies on ATS fairness analyze demographic disparities and algorithmic bias in filtering workflows [29,58]. Other works investigate entity extraction and structured parsing from resume text [14,33]. However, comparatively little research explores candidate-centric feedback systems that help job seekers interpret job requirements and systematically improve their resumes. In other words, existing literature examines how ATS systems filter candidates—not how candidates can strategically adapt their materials in response.

Our Chrome Extension takes the opposite approach. Rather than evaluating candidates for employers, we focus on empowering applicants by providing LLM-driven, personalized, and context-aware guidance. By integrating job-description analysis, resume parsing, skill-gap detection, and STAR-based rewriting into a unified pipeline, our system goes beyond traditional keyword matching [7,56]. The LLM enables narrative enhancement, improved achievement framing, clearer articulation of responsibilities, and incorporation of quantifiable impact—all factors not captured by classical ATS metrics. Additionally, the system generates recruiter-ready outreach messages and role-specific interview preparation content, bridging a major functionality gap left by commercial ATS optimization tools [42,63].

Because the extension operates directly within the user's browsing environment (e.g., LinkedIn job postings), feedback is real-time, contextual, and job-specific, reducing friction in the application workflow. This approach positions our system as a novel, candidate-oriented alternative to conventional ATS tools—one that leverages the reasoning capabilities of LLMs to provide structured, high-quality guidance rather than simple keyword suggestions.

2.3 Browser-Based ML Assistants and Chrome Extensions

Browser extensions have emerged as an effective interface for delivering machine learning assistance directly within users' browsing environments. By operating in context, these tools allow users to apply AI-driven features—such as grammar correction, content rewriting, summarization, and email drafting—without navigating away from the webpages they are viewing. This lightweight, embedded delivery model reduces friction and integrates AI support naturally into existing workflows.

Despite the rapid growth of browser-based AI tools, their use in job-application workflows remains limited. Most existing extensions specialize in narrow tasks such as rewriting LinkedIn text or generating template-based cover letters. They typically do not provide comprehensive support for analyzing job descriptions, tailoring resumes, identifying skill gaps, or preparing recruiter outreach and interview responses. Furthermore, many tools lack deep integration with job portals, requiring users to manually copy and paste job information—an inconvenience that discourages frequent tailoring of application materials.

JobAI addresses these limitations by positioning the Chrome Extension as the central interaction point for the entire job-application workflow. The extension automatically extracts job descriptions from platforms like LinkedIn, offers an interface for uploading resumes, and enables real-time LLM-powered analysis without requiring the user to leave the job posting. This context-aware design minimizes effort and encourages iterative refinement during active job searches.

Unlike single-step prompt wrappers, JobAI incorporates a multi-stage processing pipeline that includes job-description parsing, resume normalization, skills extraction, skills-gap analysis, STAR-based rewriting, and generation of recruiter messages and interview questions. By combining structured extraction with deep language reasoning, JobAI transforms resume optimization from simple keyword matching into a guided, narrative-enhancing workflow that improves clarity, alignment, and overall candidate presentation.

3 Context and Problem Definition

3.1 User Pain Points

Interviews with students and early-career professionals reveal a consistent set of challenges in navigating the job-application process. Many candidates struggle to interpret job descriptions and translate vague or broad requirements into concrete resume improvements. Even when they understand the skills listed in a posting, they are often unsure how to demonstrate those skills effectively or how to prioritize relevant experiences. A common difficulty lies in crafting strong resume bullet points—most applicants describe responsibilities rather than achievements, omit measurable outcomes, and fail to communicate impact in a structured, compelling way.

Job portals such as LinkedIn, Indeed, and Glassdoor introduce additional complexity by presenting a high volume of postings, each with slightly different expectations, terminology, and role definitions. For inexperienced applicants, distinguishing between similar roles or identifying the core competencies required can be overwhelming. Without a clear strategy, many candidates resort to applying broadly with the same generic resume, hoping that quantity will compensate for lack of tailoring.

This often leads to repeated rejections with no explanatory feedback. Because employer-side Applicant Tracking Systems do not disclose scoring criteria or alignment metrics, applicants are left guessing whether they were rejected due to missing keywords, poor resume structure, insufficient experience, or simple competition. The absence of actionable feedback amplifies frustration, reduces confidence, and discourages targeted learning. As a result, candidates lack the guidance needed to iteratively refine their application materials, identify gaps in their presentation, and improve their chances across successive job applications.

3.2 System Goals and Scope

Interviews The primary goal of our system is to function as an intelligent, candidate-centered assistant that bridges the gap between job postings and applicant preparedness. Rather than requiring job seekers to manually interpret complex role descriptions, the system analyzes job requirements and translates them into concrete, actionable guidance that users can apply directly to their resumes and interview preparation. The assistant is designed to operate in real time within the job-search environment, helping users understand what a posting truly expects and how their existing experience aligns—or fails to align—with those expectations.

The scope of the Chrome Extension encompasses a full end-to-end support workflow:

1. **Job Description (JD) Analysis**: Automatically extract and interpret job requirements from platforms such as LinkedIn to identify core skills,

responsibilities, and expectations.

2. **Resume Analysis**: Parse the user's existing resume to evaluate structure, content, quantified achievements, and alignment with the target role.

3. **Skill-Gap Identification**: Compare JD and resume content to highlight missing competencies, underrepresented skills, or areas requiring stronger articulation.

4. **Improvement Suggestions**: Provide role-specific recommendations and rewrite existing bullet points using frameworks such as STAR (Situation–Task–Action–Result).

5. **Recruiter Messaging**: Generate personalized outreach messages tailored to the job posting, improving networking and communication effectiveness.

6. **Interview Preparation**: Produce targeted technical and behavioral questions based on job responsibilities, helping candidates prepare strategically.

Importantly, the system is designed to **assist candidates**, not evaluate or rank them. It does not attempt to make hiring decisions, score applicants, or predict job-offer outcomes. Instead, the extension focuses exclusively on providing transparent and actionable guidance that empowers users to refine their presentation and better communicate their qualifications.

The current version of the system assumes that the user has at least one existing resume and a target job description. The tool does not generate complete resumes from scratch for candidates with no prior materials, although automated resume generation is a potential future enhancement. By limiting scope in this way, the system ensures high-quality, context-aware feedback while maintaining reliability and interpretability across different job roles and experience levels

3.3 Assumptions and Non-Goals

Our system operates under several key assumptions regarding user behavior and input quality. First, we assume that users provide truthful and accurate information in their resumes. The guidance generated by the system—including rewritten bullet points, STAR-based expansions, and recruiter outreach messages—is intended to refine, clarify, and strategically highlight genuine experiences rather than fabricate or inflate credentials. The extension does not promote dishonest practices or encourage users to misrepresent qualifications. Instead, it focuses on improving articulation, narrative strength, and alignment with job requirements based on the information already present.

We further assume that users possess at least one existing resume and a target job description. The system is not designed to create complete resumes from scratch for candidates with no prior experience or documented background. Similarly, it does not independently verify the authenticity of uploaded documents or cross-check user claims with external sources. The responsibility for honesty and accuracy remains with the user.

In defining the system's scope, several **non-goals** are explicitly established to avoid unintended interpretations of the system's capabilities. The tool does not offer any form of **legal, regulatory**, or **compliance** advice. Hiring practices, employment laws, data-protection rules, and anti-discrimination standards vary widely across jurisdictions and industries; providing authoritative guidance in these areas would require domain expertise, jurisdiction-specific databases, and ongoing regulatory updates that fall outside the system's intended purpose.

Additionally, while the extension aims to enhance resume quality and improve alignment with commonly used Applicant Tracking Systems, it does not attempt to reverse-engineer or precisely replicate the scoring functions of proprietary ATS platforms. These systems often rely on confidential rules, custom ranking algorithms, and employer-specific configurations. As such, JobAI focuses on general best practices—improving clarity, structure, keyword relevance, and impact articulation—rather than promising accurate prediction of ATS outcomes.

Finally, the system does not seek to automate or replace human judgment in hiring. It does not rank candidates, assess employability, or provide recommendations about job fit beyond highlighting textual alignment. Its goal is strictly to empower job seekers with clearer communication tools, actionable improvements, and better-prepared materials—not to replace career counselors, hiring managers, or professional reviewers.

4 System Design and Methodology

4.1 High-Level Architecture

The overall architecture follows a client–server model. The Chrome Extension acts as the client layer, providing user interaction, resume upload, and job description capture. The backend, implemented in Python, hosts the resume parser, prompt generator, and the Qwen2.5-Coder-7B-Instruct model. The model is served on an NVIDIA A100 20GB MIG GPU VM to support efficient inference.

When the user initiates an analysis, the extension packages the job description text and the uploaded resume PDF and sends them to the backend via a secure HTTPS API. The backend processes these inputs and returns a structured report that is displayed to the user and optionally saved as a DOCX file.

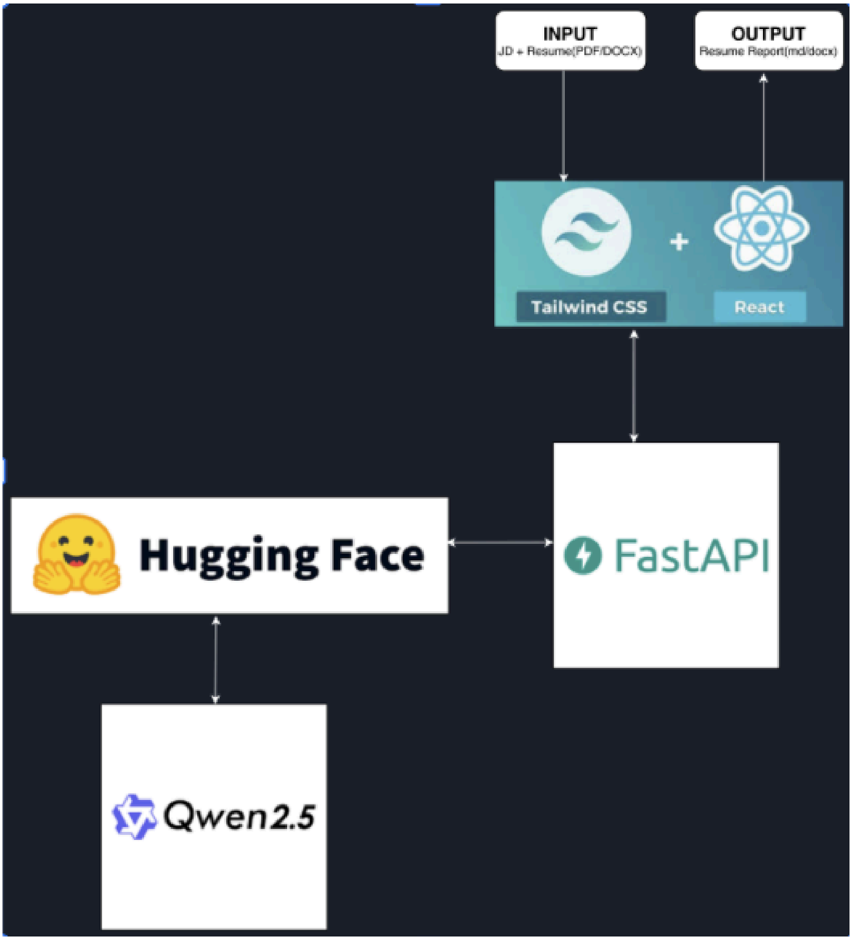

Figure 1(a): High-Level System Architecture

4.2 Backend processing pipeline

The backend pipeline consists of several modules connected in sequence: (1) resume parsing, (2) text normalization, (3) prompt construction, (4) LLM inference, and (5) report generation. The resume parser uses a PDF extraction utility to convert the candidate's resume into plain text while preserving basic line structure. The job description is taken as raw text from the Chrome Extension and normalized to remove extra whitespace and non-printable characters.

The prompt construction module combines a role description, a detailed task description, and the cleaned JD and resume text into a single structured prompt. This prompt is passed to the Qwen2.5-Coder-7B-Instruct model using a chat-style interface. The model generates a long-form response that is post-processed into sections, which are then written into a DOCX document for download.

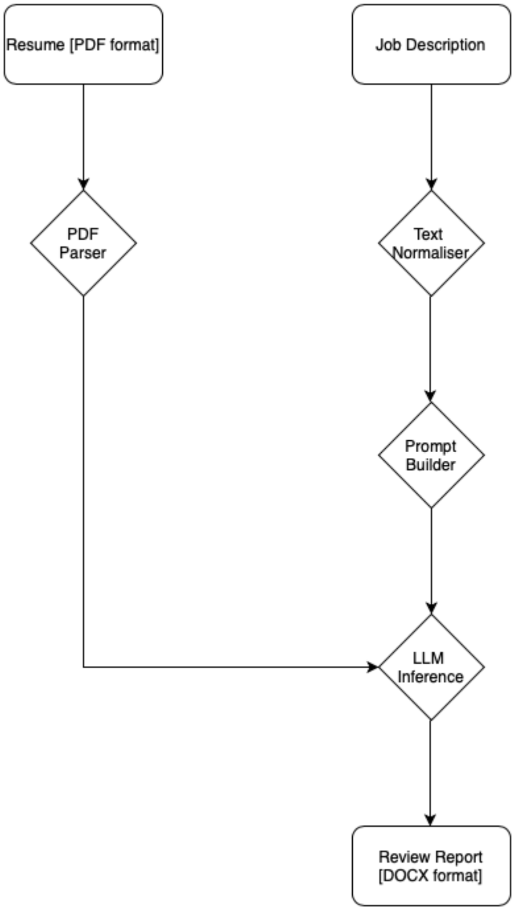

Figure 2(a): Backend Processing Pipeline

4.3 Prompt Engineering Strategy

Prompt engineering is central to the system's performance. Rather than issuing free-form instructions, we define a strict template that enumerates all desired sections in a fixed order: key JD skills, skills found in the resume, missing or weak skills, recommended resume additions, new STAR bullet points, recruiter cold email, LinkedIn outreach message, resume score breakdown, and interview preparation Q&A.

The system prompt describes the model's role as an ATS optimization assistant and interview coach. The user prompt concatenates the instructions with the normalized job description and resume. Temperature is kept low (e.g., 0.15–0.2) to favor determinism and reduce hallucinations. The maximum token limit is chosen to ensure that the entire report can be generated in a single pass without truncation.

5 Experiments and Results

5.1 Experimental Setup

We evaluate the system using a set of realistic resume–JD pairs collected from students and public job postings. For each pair, we generate a full report using the Chrome Extension and the backend pipeline. We focus on qualitative dimensions such as clarity, relevance, and usefulness of suggestions, as well as system-level metrics such as latency and robustness to variation in input quality.

All experiments are run on a single NVIDIA A100 20GB MIG GPU VM. Average end-to-end response time, including PDF parsing, prompt construction, LLM inference, and DOCX generation, is approximately 10–18 seconds for typical resume lengths (1–2 pages) and moderately sized job descriptions.

5.2 Qualitative Case Study Results

In one case study, we analyze a candidate resume oriented primarily around teaching assistance and tutoring roles against a back-end developer job description. The system successfully identifies that the candidate possesses relevant programming and database skills, but lacks explicit back-end project experience and API implementation examples. The generated recommendations suggest highlighting any relevant back-end coursework, side projects, or internships and rewriting bullets to emphasize implementation, optimization, and collaboration.

The STAR-style bullet points generated by the system introduce concrete metrics and outcomes, such as reduced query execution time or improved reliability of a system. Evaluators reported that these bullets read similarly to those found on resumes of experienced professionals and clearly communicated impact.

Recruiter emails and LinkedIn outreach messages produced by the system were judged to be professional, concise, and appropriately personalized to the company and role. Interview questions were aligned with the job description and resume content, covering both technical topics and behavioral scenarios.

5.3 Output File Structure and Analysis

We analyze the structure of a sample output report generated by the system. The report adheres strictly to the template defined in the prompt and is divided into logically separated sections. The first section lists key skills required by the job description, grouping them into technical skills, tools, responsibilities, and soft skills. The next section mirrors this grouping for skills found in the resume, enabling easy side-by-side comparison.

The gap analysis section enumerates missing or weak skills, such as lack of explicit API design or absence of back-end optimization examples. Recommended resume additions translate these gaps into actionable suggestions—for instance, asking the candidate to describe server-side responsibilities in more detail or to quantify the effect of process improvements.

The STAR bullet section generates new, high-quality resume lines with measurable outcomes (e.g., reducing manual checks by 30 percent, improving query performance by 50 percent). This section is particularly valuable for candidates who struggle with phrasing achievements. Following this, the system generates a recruiter cold email and LinkedIn outreach message tailored to the company and role.

The resume score breakdown assigns separate scores for skill match, experience alignment, impact and metrics, and ATS keyword coverage, along with an overall score. Explanatory text clarifies why each score was assigned, helping candidates prioritize improvement areas. Finally, the interview Q&A section provides technical, behavioral, and resume-deep-dive questions with sample answers, serving as a compact study guide.

6 Discussion

6.1 Strengths

The primary strength of the system lies in its integration of multiple job-application tasks into a single workflow. Instead of using separate tools for resume review, email drafting, and interview preparation, candidates can obtain all three from one interaction. The Chrome Extension interface lowers friction by appearing directly on job posting pages. Another strength is the structured nature of the generated reports. Because the output format is tightly controlled by the prompt template, candidates receive consistent guidance across different roles and resumes. The mixture of qualitative explanations, quantitative scores, and concrete examples makes the feedback more actionable than simple keyword lists.

6.2 Limitations

Despite its advantages, the system has several limitations. First, the evaluation is primarily qualitative and based on a limited number of case studies. A larger-scale user study involving actual job outcomes would be required to rigorously measure impact on interview callbacks and offer rates.

Second, the system currently relies on a single LLM and does not dynamically adapt to different sectors (e.g., healthcare vs. finance). While the prompt template is generic, certain roles may require specialized knowledge or jargon that the base model does not fully capture.

Third, the system does not generate a fully rewritten resume document. Instead, it generates suggestions and example bullet points that users must manually incorporate. While this is a deliberate choice to maintain user control and avoid hallucinating experience, some users may prefer a more automated rewriting mode.

6.3 Ethical Considerations

Using LLMs in hiring-related contexts raises ethical concerns. Although our system is designed to assist candidates rather than employers, there is still a risk that generated content could encourage exaggeration or misrepresentation if used irresponsibly. We mitigate this risk through instructions that emphasize staying truthful and realistic.

Bias is another potential issue. LLMs are trained on large corpora that may reflect societal biases. While our use case focuses on phrasing and alignment rather than ranking candidates, it remains important to monitor for biased language or suggestions. Future work may include bias detection and mitigation modules to audit generated content.

7 Future Work

Several directions for future enhancement emerge from our current design. From the beginning, our goal has been to support SFSU students and provide them with a practical tool that we could eventually hand off to the Career Development Department. With access to anonymized data from the department—such as interview questions, hiring patterns, and strong resume samples—we can feed this information into a Retrieval-Augmented Generation (RAG) system and use that enriched knowledge base to produce more accurate and context-aware reports for students.

Another extension is automated resume rewriting. Based on the current report structure, the system could apply modifications directly to a copy of the user's resume, reordering sections, inserting suggested bullet points, and refining language to improve clarity. This capability would require strong safeguards to avoid introducing errors or altering factual information.

Finally, reinforcement learning could be used to close the loop between suggestions and real outcomes. With user consent, the system could track which recommendations are accepted and which applications result in interviews or offers, allowing the model to adjust scoring heuristics and prompt configurations to aim for stronger ATS alignment and better recruiter engagement.

8 Conclusion

We presented an AI-Powered Job Application Chrome Extension that leverages a large language model to analyze job descriptions and resumes, identify skill gaps, generate STAR-based bullet points, draft recruiter outreach messages, and prepare candidates for interviews. By embedding this functionality inside a browser extension, we reduce friction and make advanced AI capabilities accessible to freshers and job seekers directly within their existing job search workflows.

Our qualitative experiments suggest that the system produces coherent, relevant, and practically useful guidance. At the same time, we recognize the need for broader evaluation, expanded domain coverage, and additional safeguards against bias and misuse. As LLMs continue to evolve, tools like this Chrome Extension have the potential to democratize access to high-quality career guidance and improve outcomes for a wide range of candidates.

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